

```
In [1]: import os
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDisplay
import tensorflow as tf
from tensorflow.keras import layers, models, callbacks
from helpers import *

np.random.seed(42)
tf.random.set_seed(42)
```

```
In [2]: DATASET_DIR = "../planetsdataset"
```

```
In [3]: IMG_SIZE = (64, 64)
        BATCH_SIZE = 32
```

```
In [4]: train_dataset, valid_dataset, test_dataset, class_names = load_datasets(DATASET_DIR)

Found 2430 files belonging to 12 classes.
Found 666 files belonging to 12 classes.
Found 341 files belonging to 12 classes.
```

```
In [5]: class_names
```

```
Out[5]: ['asteroid',
         'earth',
         'galaxy',
         'jupiter',
         'mars',
         'mercury',
         'moon',
         'neptune',
         'pluto',
         'saturn',
         'uranus',
         'venus']
```

```
In [6]: num_classes = len(class_names)
```

```
In [7]: model = models.Sequential([
        layers.Input(shape=(IMG_SIZE[0], IMG_SIZE[1], 3)),
        layers.Rescaling(1.0 / 255),
        layers.Flatten(),
        layers.Dense(128, activation="relu"),
        layers.Dense(num_classes, activation="softmax")
    ])

model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)
```

```
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 64, 64, 3)	0
flatten (Flatten)	(None, 12288)	0
dense (Dense)	(None, 128)	1572992
dense_1 (Dense)	(None, 12)	1548

=====
 Total params: 1574540 (6.01 MB)
 Trainable params: 1574540 (6.01 MB)
 Non-trainable params: 0 (0.00 Byte)

```
In [8]: early_stopping = callbacks.EarlyStopping(
        monitor="val_loss",
        patience=5,
        restore_best_weights=True
    )
```

```
In [9]: history = model.fit(
        train_dataset,
        validation_data=valid_dataset,
        epochs=20,
        callbacks=[early_stopping]
    )
```

Epoch 1/20

55/76 [=====>.....] - ETA: 2s - loss: 2.4284 - accuracy: 0.3574

2026-04-19 20:52:23.040678: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 11s 127ms/step - loss: 2.1709 - accuracy: 0.4095 - val_loss: 1.2930 - val_accuracy: 0.5721

Epoch 2/20

55/76 [=====>.....] - ETA: 2s - loss: 1.1555 - accuracy: 0.6392

2026-04-19 20:52:33.538821: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 11s 132ms/step - loss: 1.1188 - accuracy: 0.6486 - val_loss: 1.0512 - val_accuracy: 0.6742

Epoch 3/20

55/76 [=====>.....] - ETA: 2s - loss: 0.9010 - accuracy: 0.7494

2026-04-19 20:52:44.879436: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 11s 131ms/step - loss: 0.8804 - accuracy: 0.7539 - val_loss: 0.8821 - val_accuracy: 0.7823

Epoch 4/20

55/76 [=====>.....] - ETA: 2s - loss: 0.7316 - accuracy: 0.7983

2026-04-19 20:52:55.330057: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 10s 117ms/step - loss: 0.7302 - accuracy: 0.7942 - val_loss: 0.8680 - val_accuracy: 0.7793

Epoch 5/20

55/76 [=====>.....] - ETA: 2s - loss: 0.6660 - accuracy: 0.8108

2026-04-19 20:53:05.863896: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 11s 127ms/step - loss: 0.6536 - accuracy: 0.8115 - val_loss: 0.9329 - val_accuracy: 0.7658

Epoch 6/20

55/76 [=====>.....] - ETA: 2s - loss: 0.5951 - accuracy: 0.8318

2026-04-19 20:53:15.970314: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 10s 126ms/step - loss: 0.5818 - accuracy: 0.8366 - val_loss: 0.6994 - val_accuracy: 0.8273

Epoch 7/20

55/76 [=====>.....] - ETA: 2s - loss: 0.4646 - accuracy: 0.8767

2026-04-19 20:53:26.536226: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 10s 119ms/step - loss: 0.4671 - accuracy: 0.8745 - val_loss: 0.7442 - val_accuracy: 0.8123

Epoch 8/20

55/76 [=====>.....] - ETA: 2s - loss: 0.4702 - accuracy: 0.8597

2026-04-19 20:53:36.083374: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 9s 113ms/step - loss: 0.4546 - accuracy: 0.8667 - val_loss: 0.7373 - val_accuracy: 0.8093

Epoch 9/20

55/76 [=====>.....] - ETA: 2s - loss: 0.3916 - accuracy: 0.8989

2026-04-19 20:53:45.524330: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 9s 113ms/step - loss: 0.3943 - accuracy: 0.8914 - val_loss: 0.6862 - val_accuracy: 0.8273

Epoch 10/20

55/76 [=====>.....] - ETA: 2s - loss: 0.3517 - accuracy: 0.9017

2026-04-19 20:53:54.927172: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 9s 112ms/step - loss: 0.3651 - accuracy: 0.8955 - val_loss: 0.6890 - val_accuracy: 0.8273

Epoch 11/20

55/76 [=====>.....] - ETA: 2s - loss: 0.3938 - accuracy: 0.8824

2026-04-19 20:54:04.328136: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 9s 112ms/step - loss: 0.3901 - accuracy: 0.8881 - val_loss: 0.6660 - val_accuracy: 0.8183

Epoch 12/20

55/76 [=====>.....] - ETA: 2s - loss: 0.2882 - accuracy: 0.9261

2026-04-19 20:54:13.705820: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 9s 112ms/step - loss: 0.2875 - accuracy: 0.9276 - val_loss: 0.7241 - val_accuracy: 0.8288

Epoch 13/20

55/76 [=====>.....] - ETA: 2s - loss: 0.2880 - accuracy: 0.9273

2026-04-19 20:54:23.103377: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 9s 112ms/step - loss: 0.2835 - accuracy: 0.9255 - val_loss: 0.6649 - val_accuracy: 0.8468

Epoch 14/20

55/76 [=====>.....] - ETA: 2s - loss: 0.2910 - accuracy: 0.9210

2026-04-19 20:54:33.092428: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 10s 118ms/step - loss: 0.2851 - accuracy: 0.9198 - val_loss: 0.6596 - val_accuracy: 0.8393

Epoch 15/20

55/76 [=====>.....] - ETA: 2s - loss: 0.2146 - accuracy: 0.9511

2026-04-19 20:54:42.576886: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 9s 112ms/step - loss: 0.2180 - accuracy: 0.9494 - val_loss: 0.6047 - val_accuracy: 0.8604

Epoch 16/20

55/76 [=====>.....] - ETA: 2s - loss: 0.1864 - accuracy: 0.9574

2026-04-19 20:54:51.978532: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 9s 112ms/step - loss: 0.2002 - accuracy: 0.9506 - val_loss: 0.6436 - val_accuracy: 0.8574

Epoch 17/20

55/76 [=====>.....] - ETA: 2s - loss: 0.1798 - accuracy: 0.9602

2026-04-19 20:55:01.413134: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 9s 112ms/step - loss: 0.1917 - accuracy: 0.9560 - val_loss: 0.7097 - val_accuracy: 0.8559

Epoch 18/20

55/76 [=====>.....] - ETA: 2s - loss: 0.1902 - accuracy: 0.9477

2026-04-19 20:55:10.801524: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 9s 112ms/step - loss: 0.1815 - accuracy: 0.9519 - val_loss: 0.6966 - val_accuracy: 0.8619
 Epoch 19/20
 55/76 [=====>.....] - ETA: 2s - loss: 0.1527 - accuracy: 0.9614

2026-04-19 20:55:20.233929: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

76/76 [=====] - 9s 112ms/step - loss: 0.1674 - accuracy: 0.9588 - val_loss: 0.6841 - val_accuracy: 0.8544
 Epoch 20/20
 55/76 [=====>.....] - ETA: 2s - loss: 0.1422 - accuracy: 0.9659

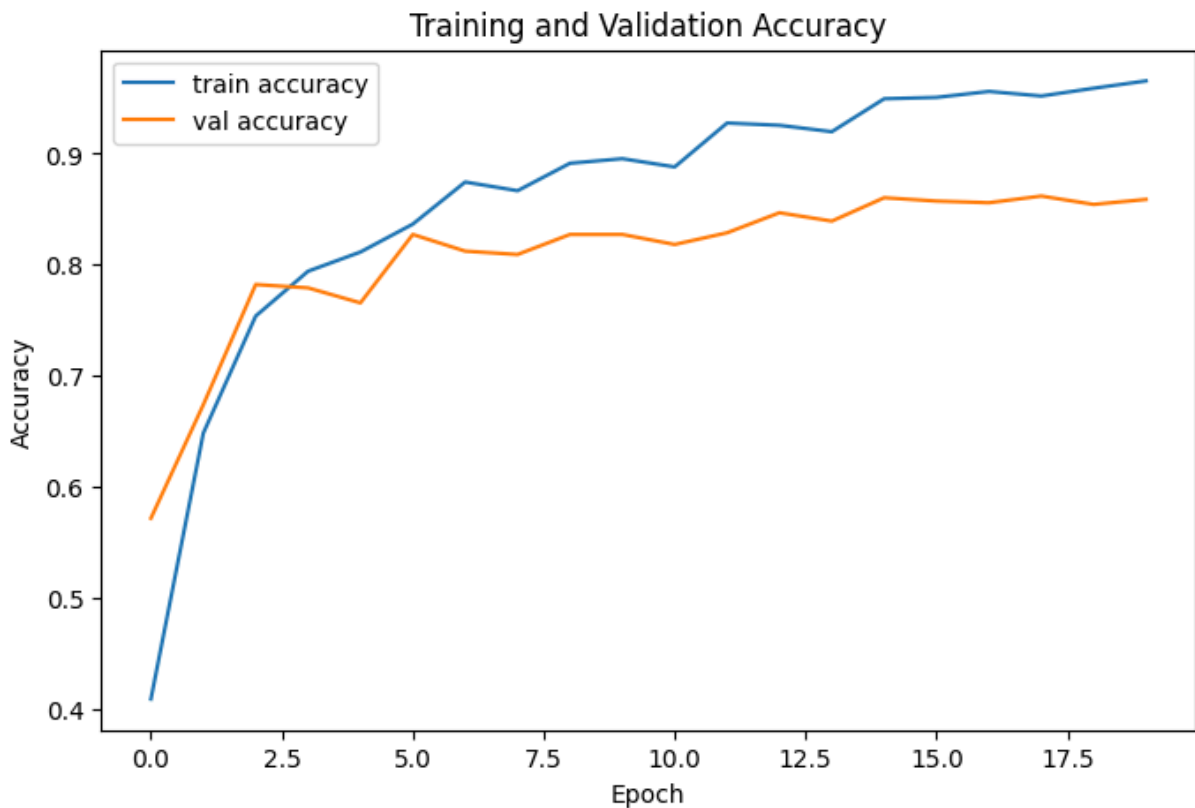
2026-04-19 20:55:29.643918: W tensorflow/core/lib/png/png_io.cc:88] PNG warning: iCCP: known incorrect sRGB profile

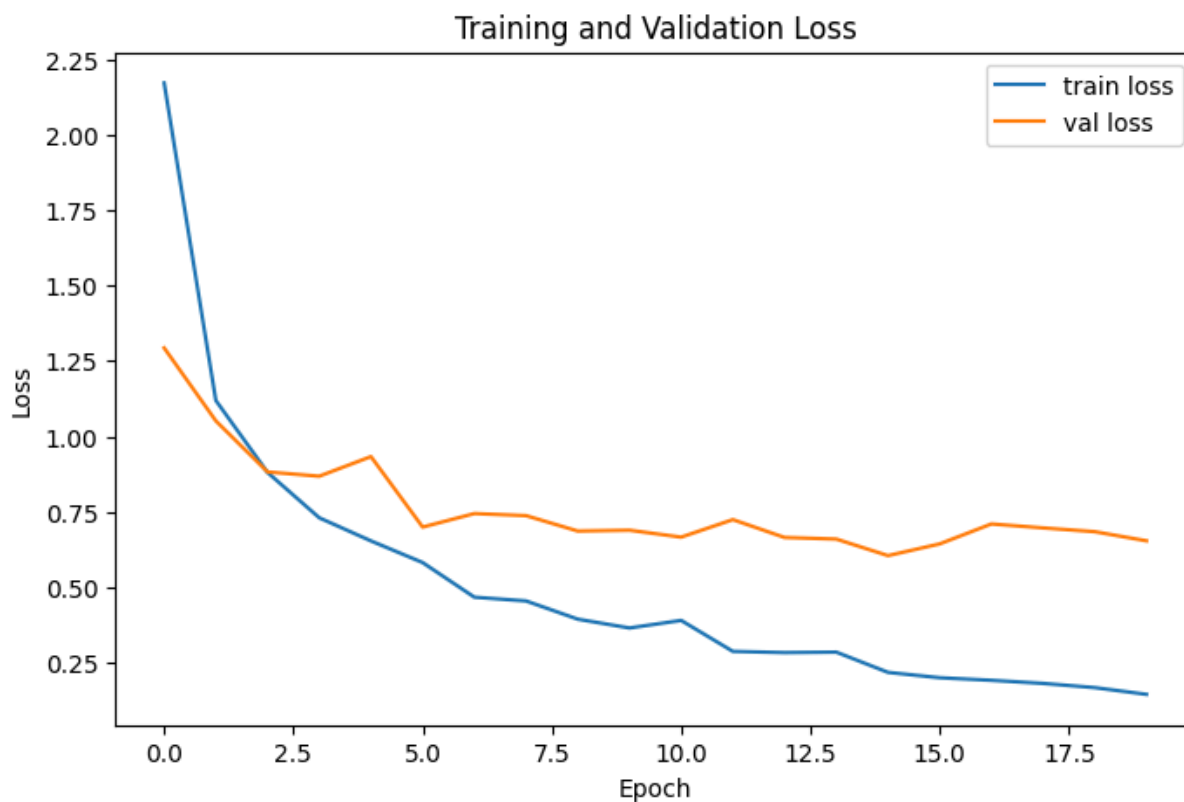
76/76 [=====] - 9s 112ms/step - loss: 0.1451 - accuracy: 0.9654 - val_loss: 0.6538 - val_accuracy: 0.8589

```
In [10]: test_loss, test_acc = model.evaluate(test_dataset)
print("Test loss:", test_loss)
print("Test accuracy:", test_acc)
```

11/11 [=====] - 1s 18ms/step - loss: 0.5647 - accuracy: 0.8622
 Test loss: 0.5646674036979675
 Test accuracy: 0.8621701002120972

```
In [11]: plot_history(history)
```





```
In [12]: y_pred_probs = model.predict(test_dataset)
y_pred = np.argmax(y_pred_probs, axis=1)
y_true = np.concatenate([y for x, y in test_dataset], axis=0)
cm = confusion_matrix(y_true, y_pred)
```

11/11 [=====] - 1s 14ms/step

```
In [13]: fig, ax = plt.subplots(figsize=(10, 10))
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=class_name)
disp.plot(ax=ax, xticks_rotation=45, cmap="Blues", colorbar=False)
plt.show()
```

